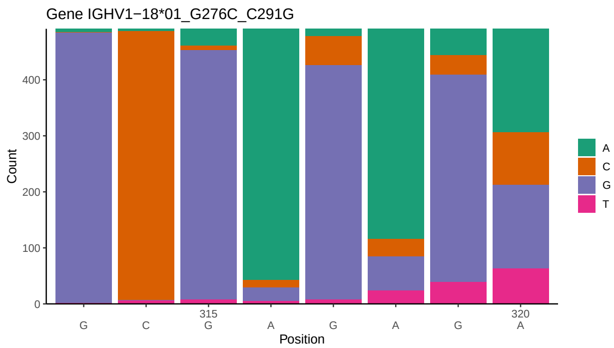
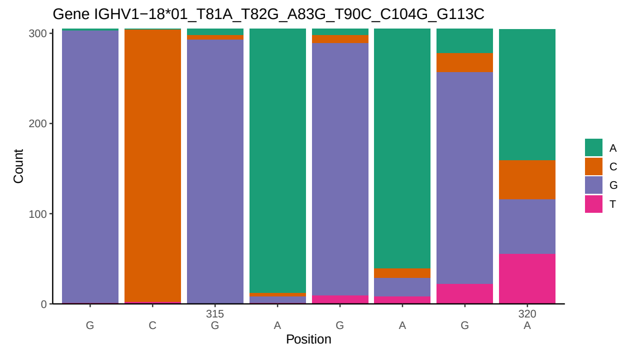
OGRDBstats Report

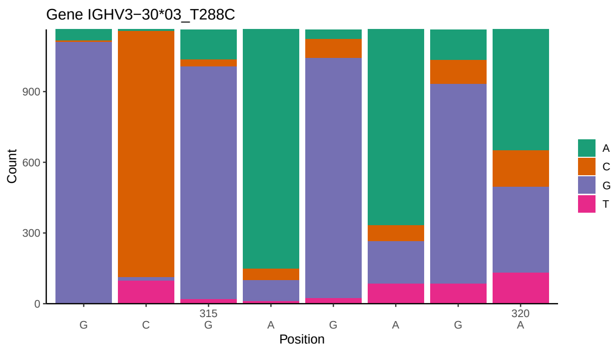
Contents

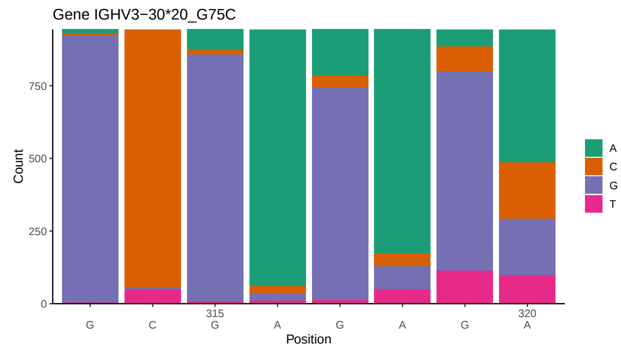
1 Novel sequence analysis

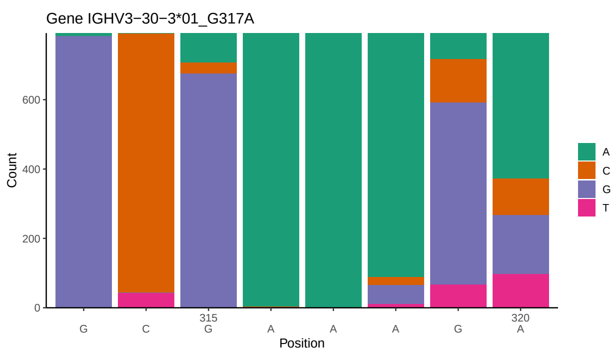
1.1 End-nucleotide composition

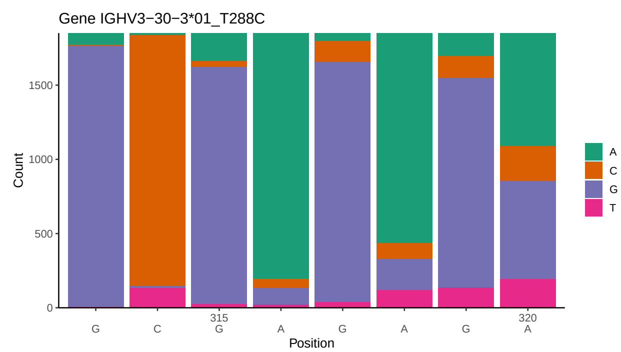


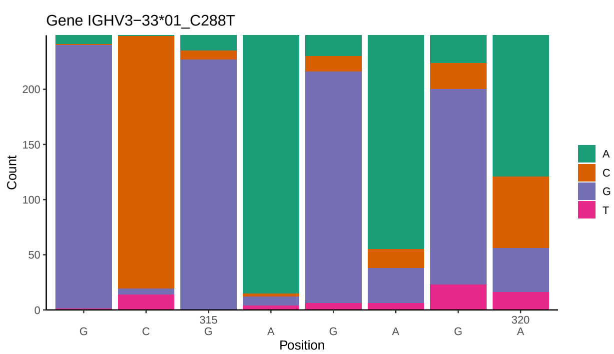


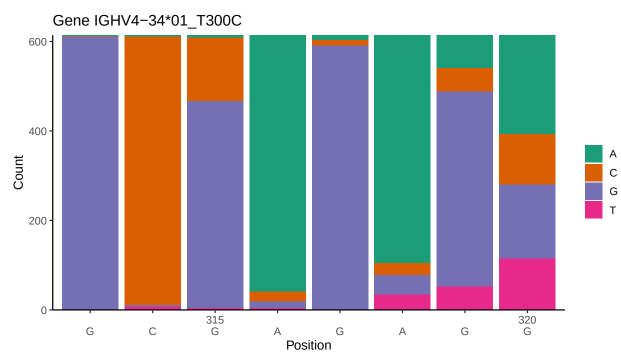


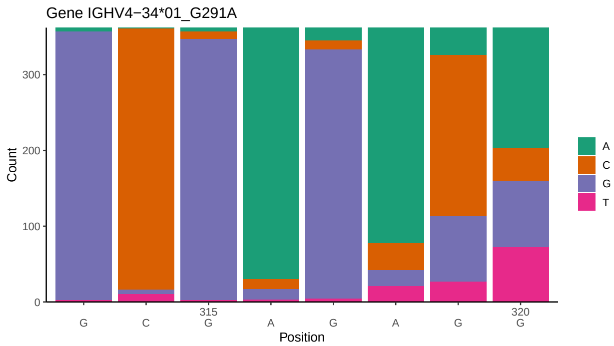


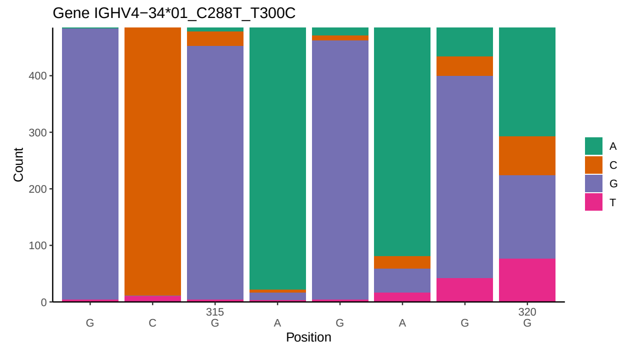


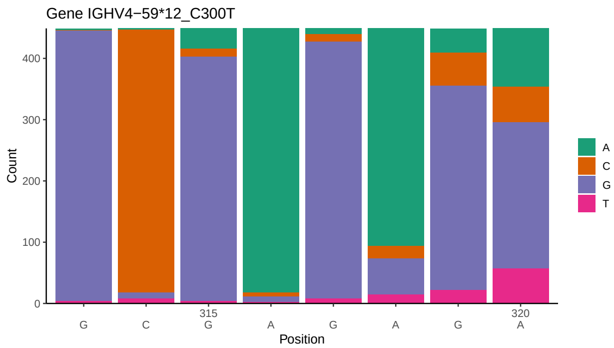




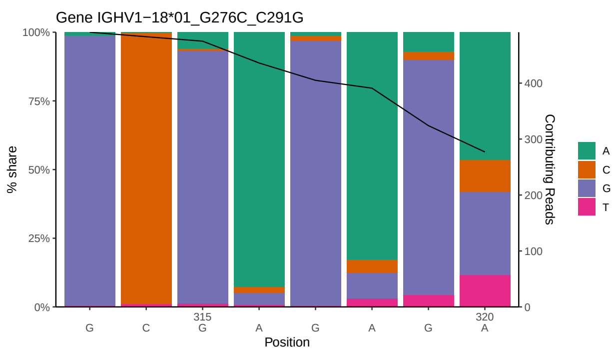


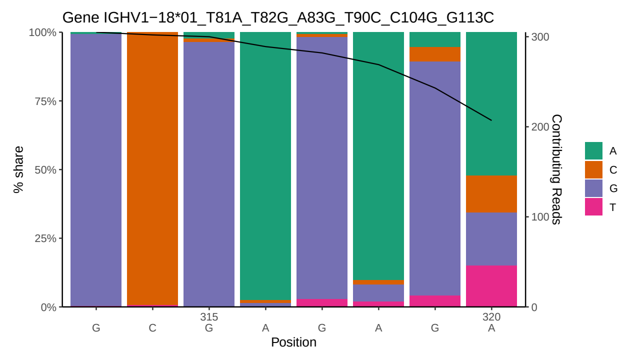


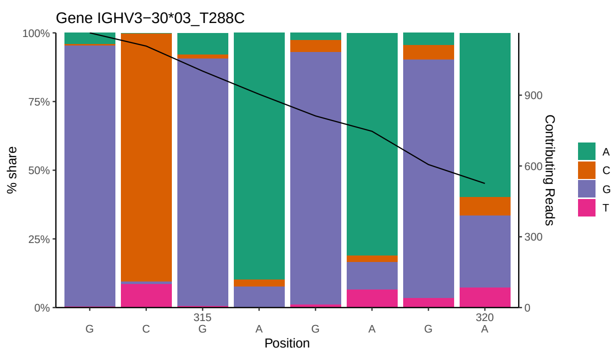


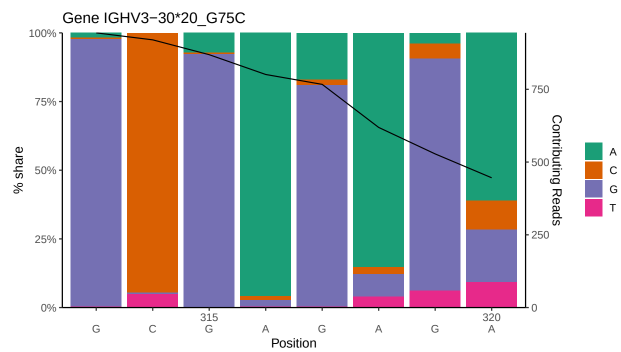


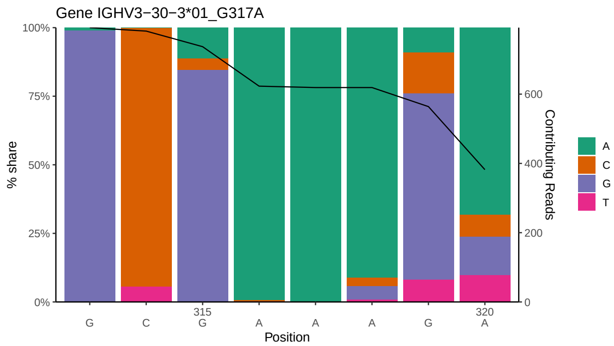
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

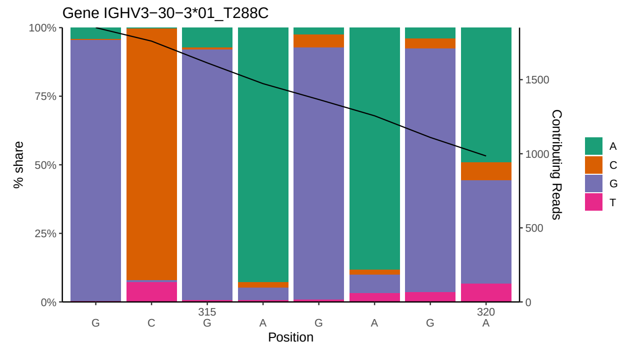


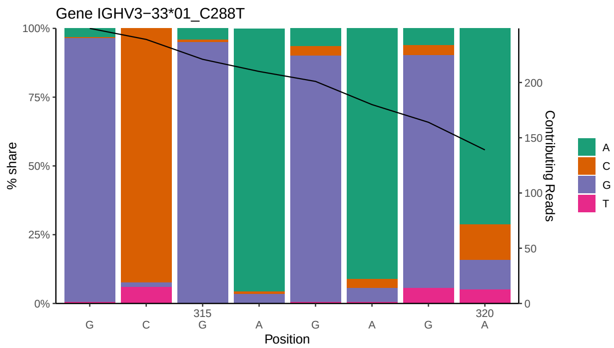


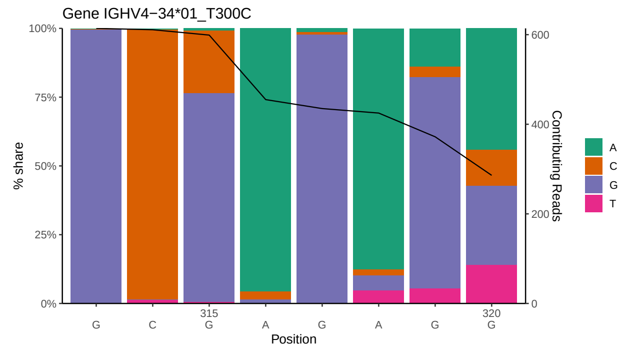


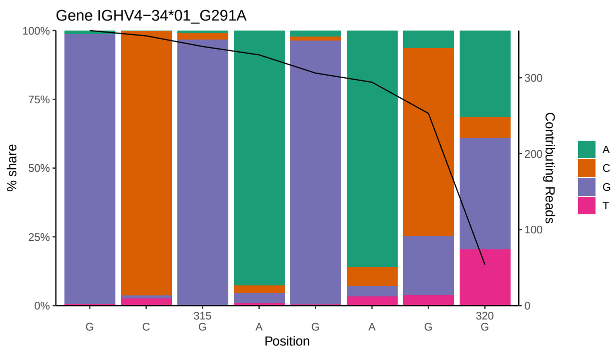


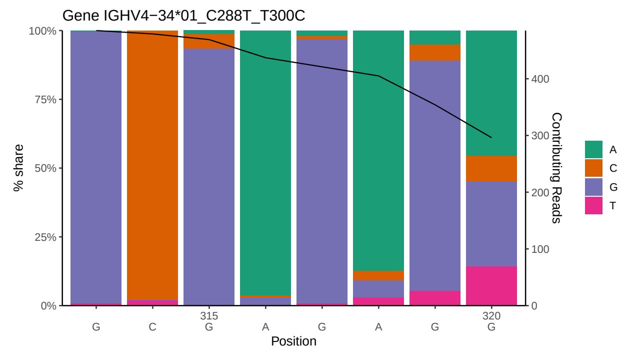

Gene IGHV3−30−3*01_G317A
% share
Contributing Reads
Position
G C G A A A G A
315 320
A
C
G
T

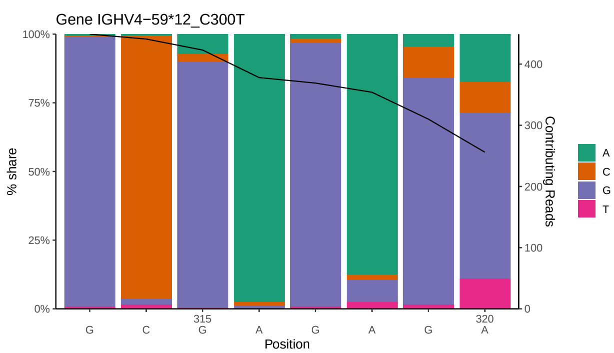

Gene IGHV3−30−3*01_T288C
% share
Contributing Reads
Position
G C G A G A G A
315 320
A
C
G
T


Gene IGHV3−33*01_C288T
% share
Contributing Reads
Position
G C G A G A G A
315 320
A
C
G
T

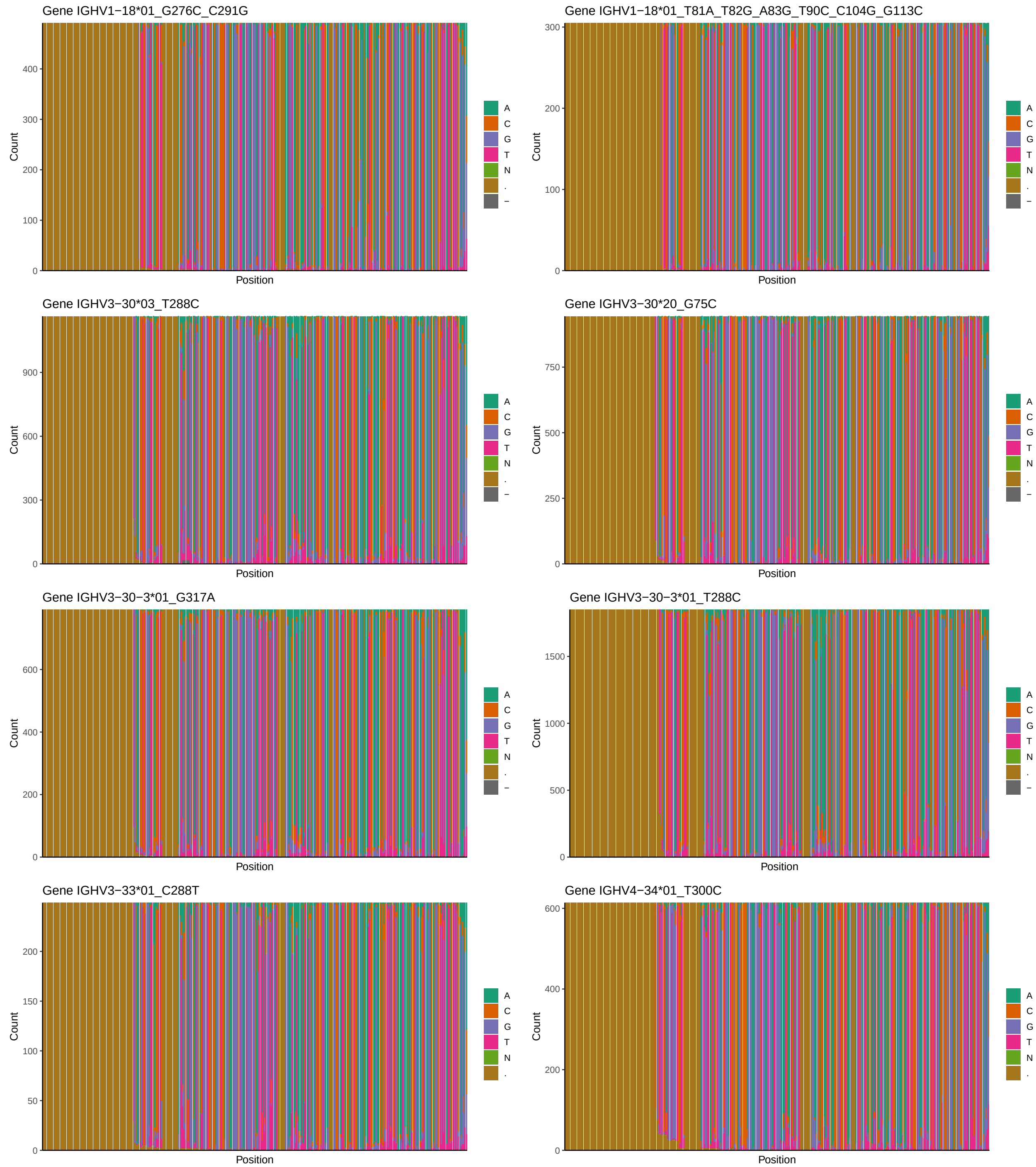

Gene IGHV4−34*01_T300C
% share
Contributing Reads
Position
G C G A G A G G
315 320
A
C
G
T

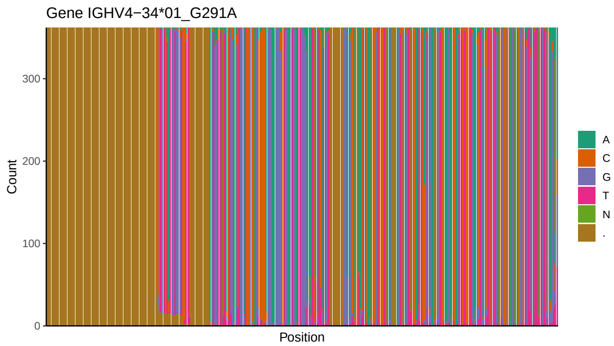

Gene IGHV4−34*01_G291A
% share
Contributing Reads
Position
G C G A G A G G
315 320
A
C
G
T

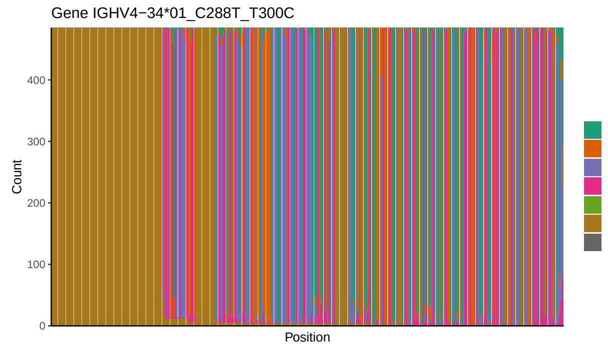

Gene IGHV4−34*01_C288T_T300C
% share
Contributing Reads
Position
G C G A G A G G
315 320
A
C
G
T

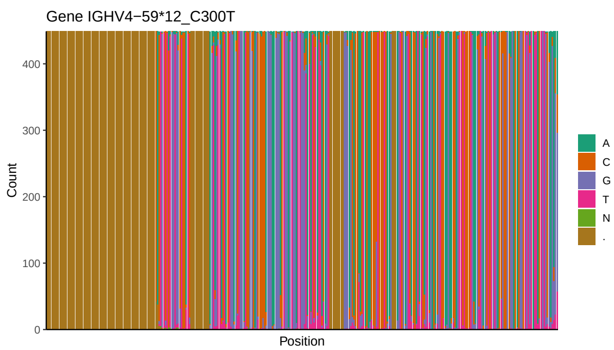

Gene IGHV4−59*12_C300T
% share
Contributing Reads
Position
G C G A G A G A
315 320
A
C
G
T

1.3 Whole-sequence composition of each assigned read

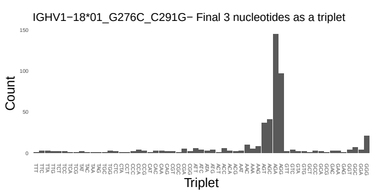


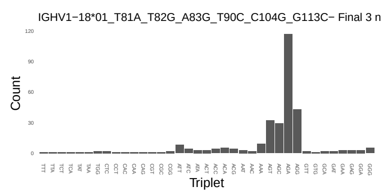


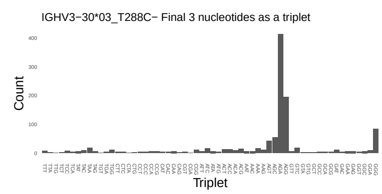


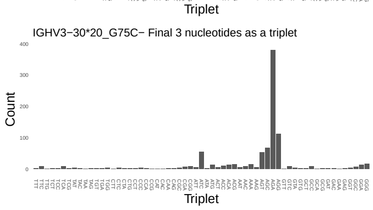


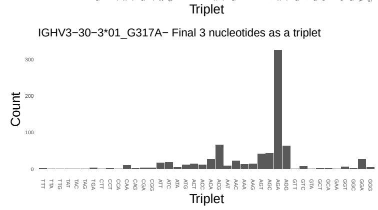
1.4 Final three nucleotides: frequency of each observed triplet

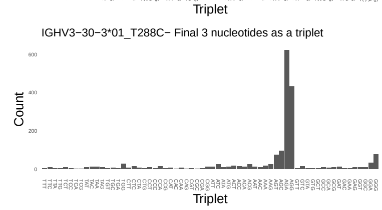


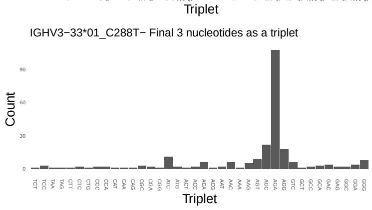


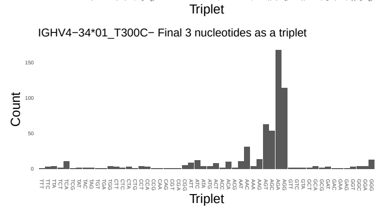


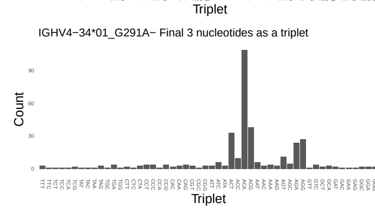


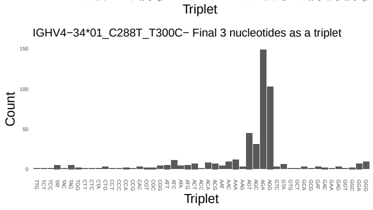


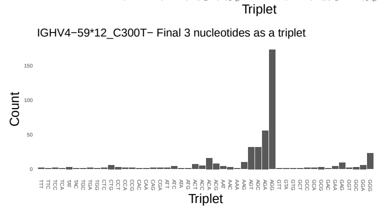




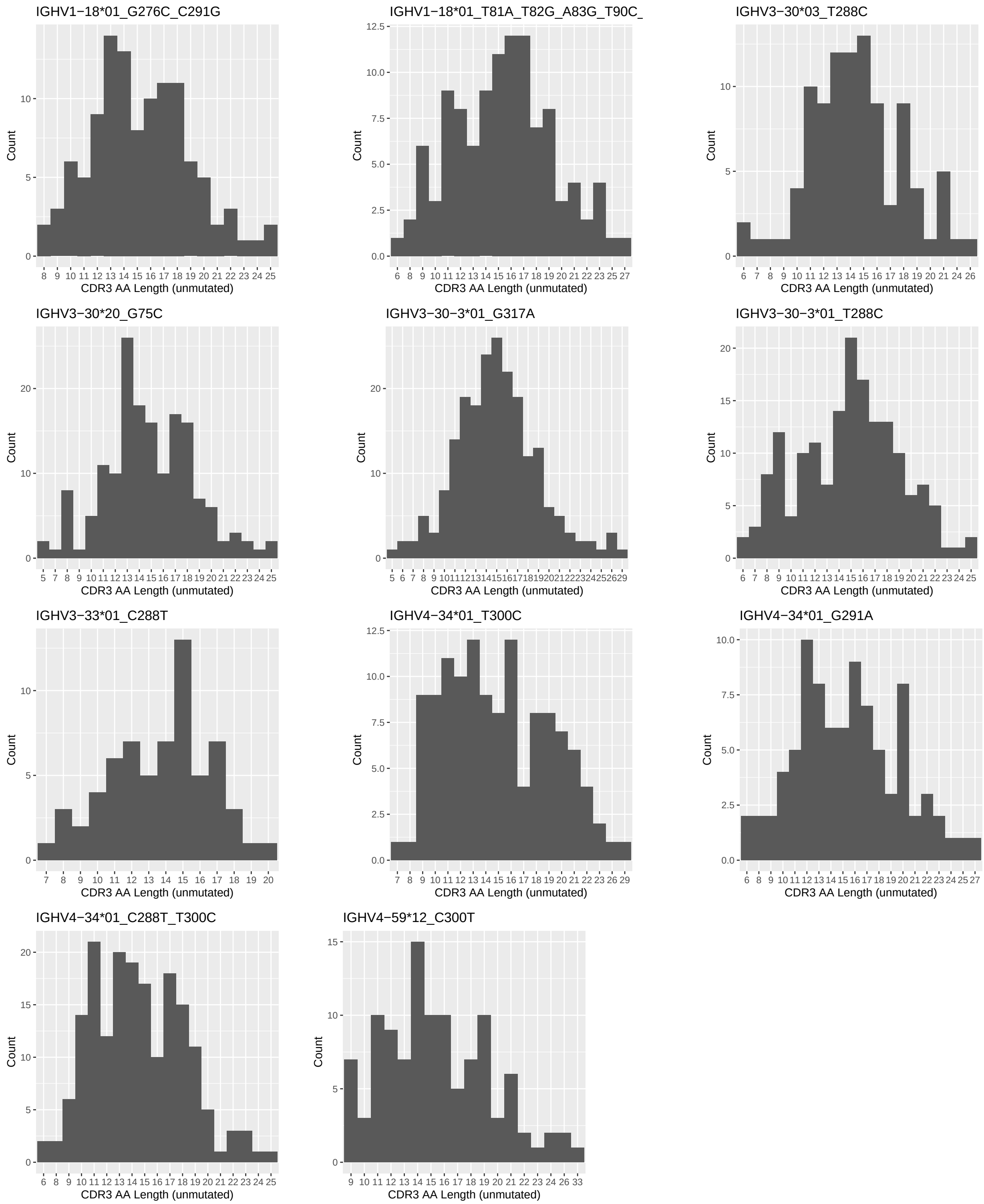




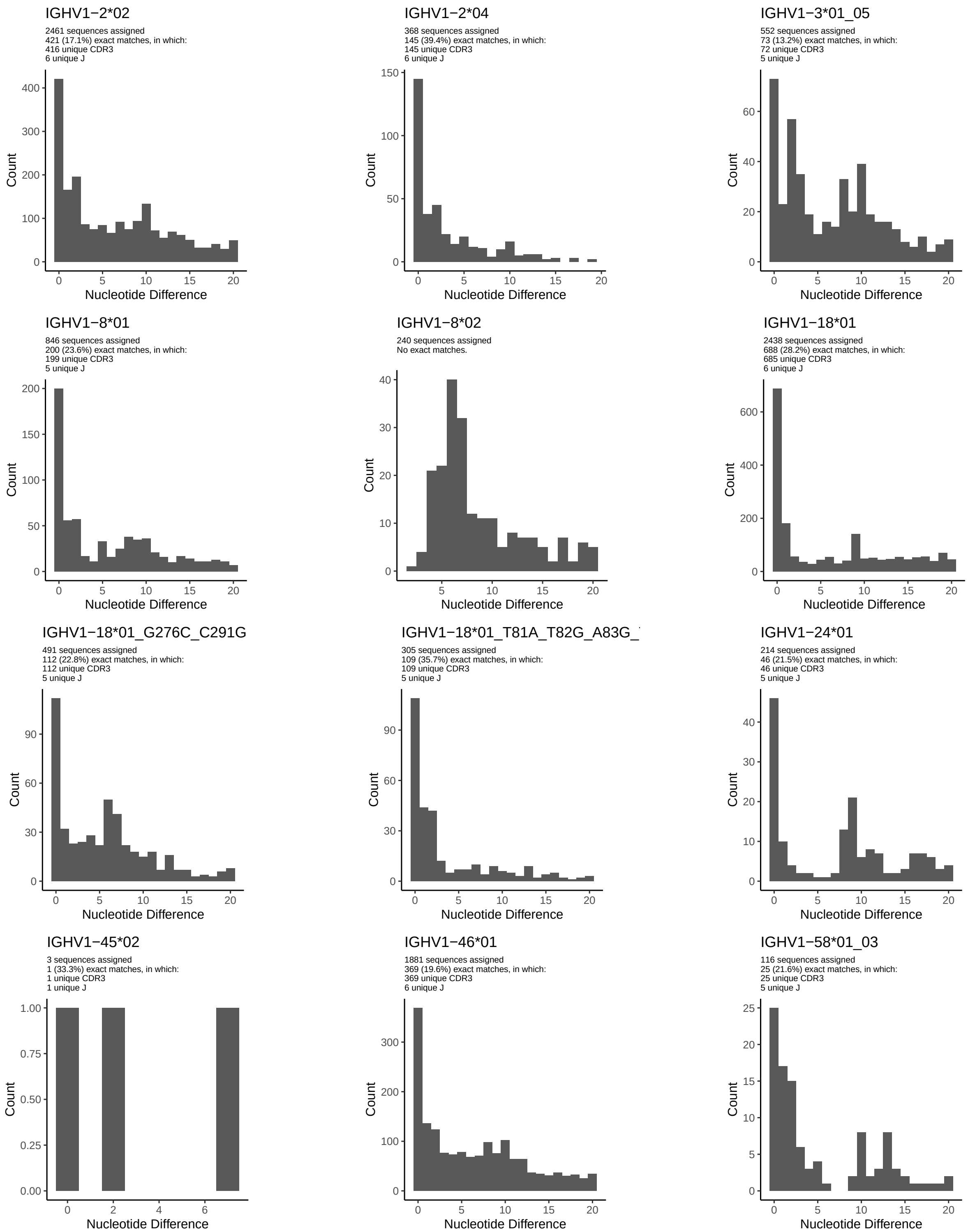


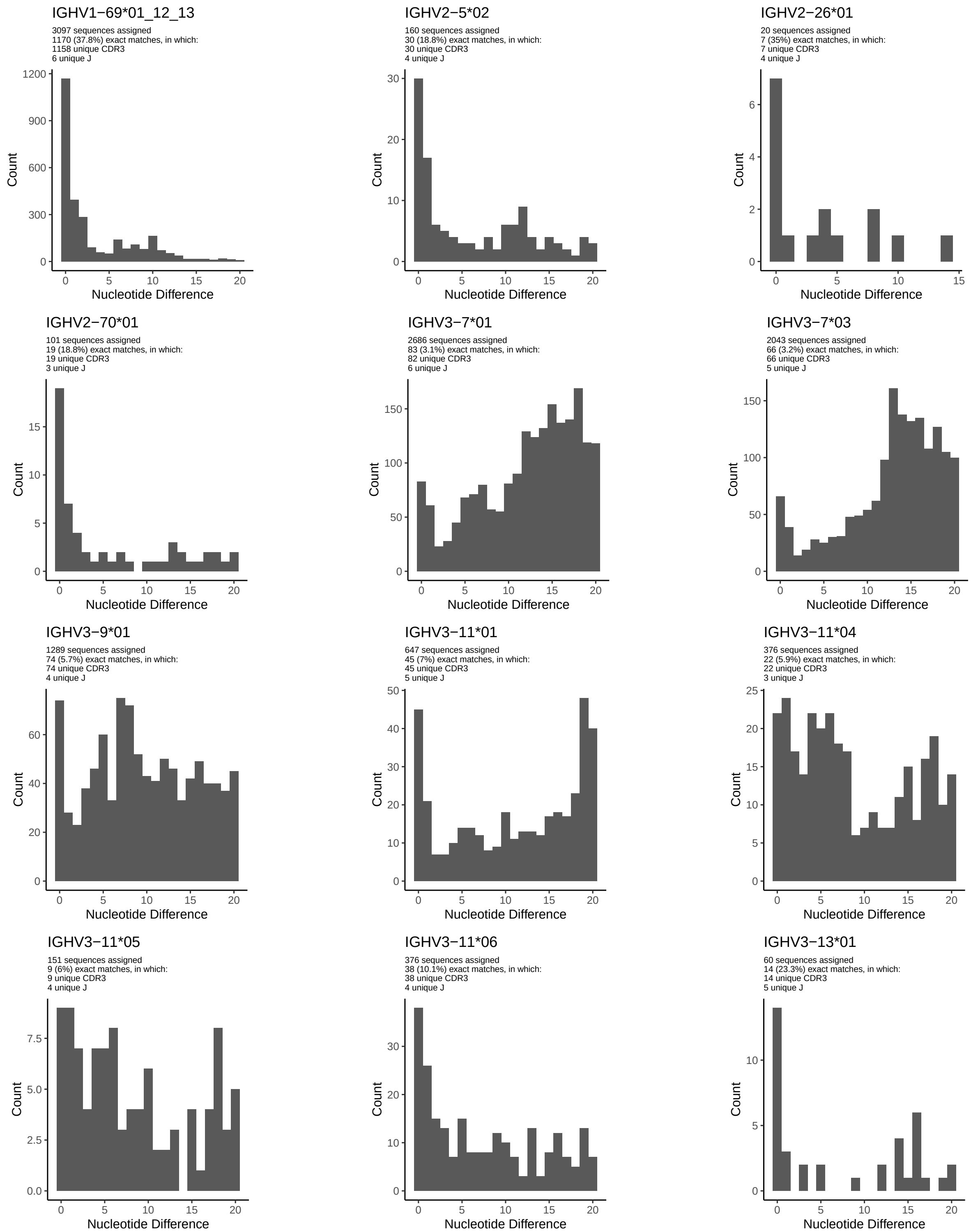


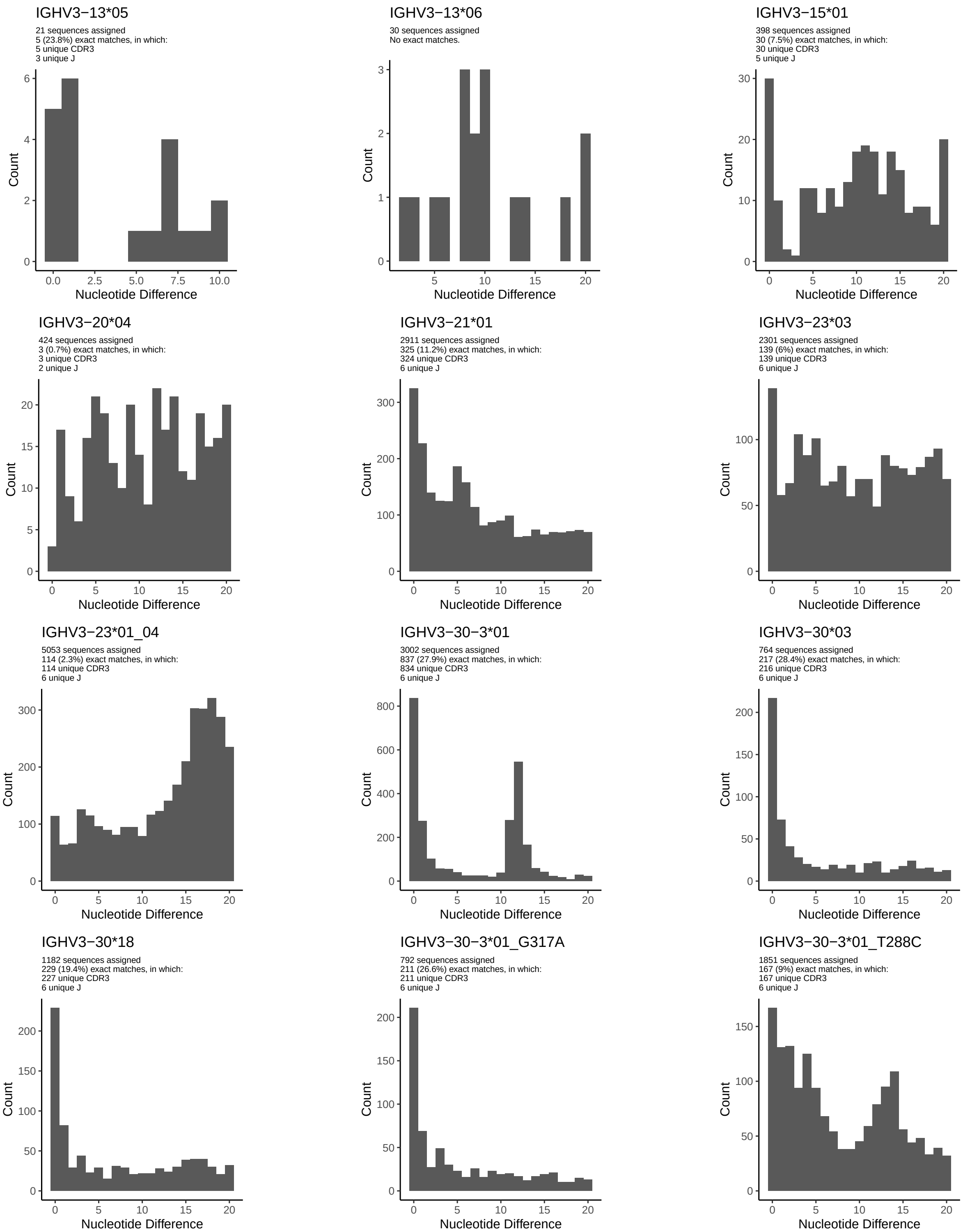
1.5 CDR3 length distribution, in assignments to novel alleles

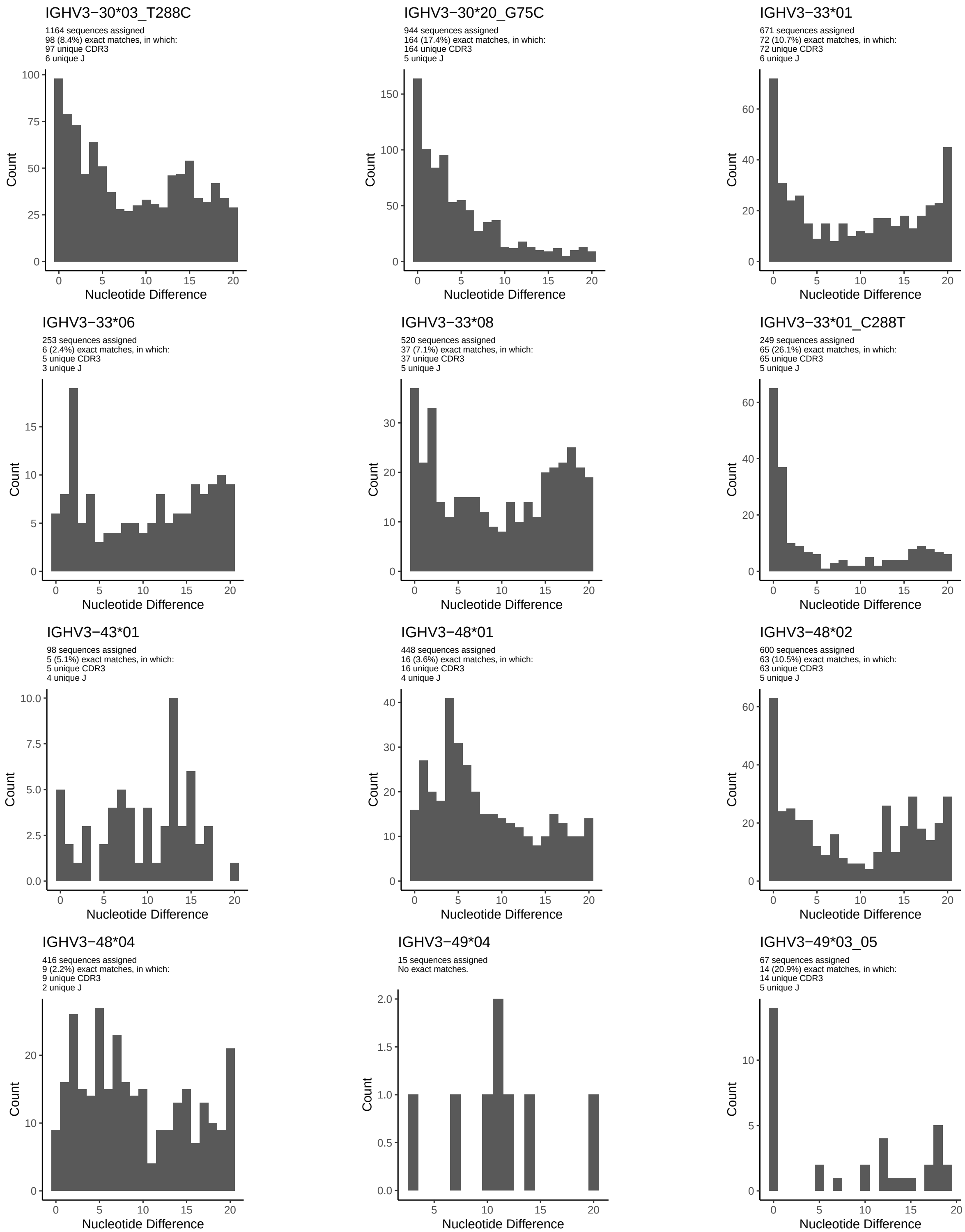


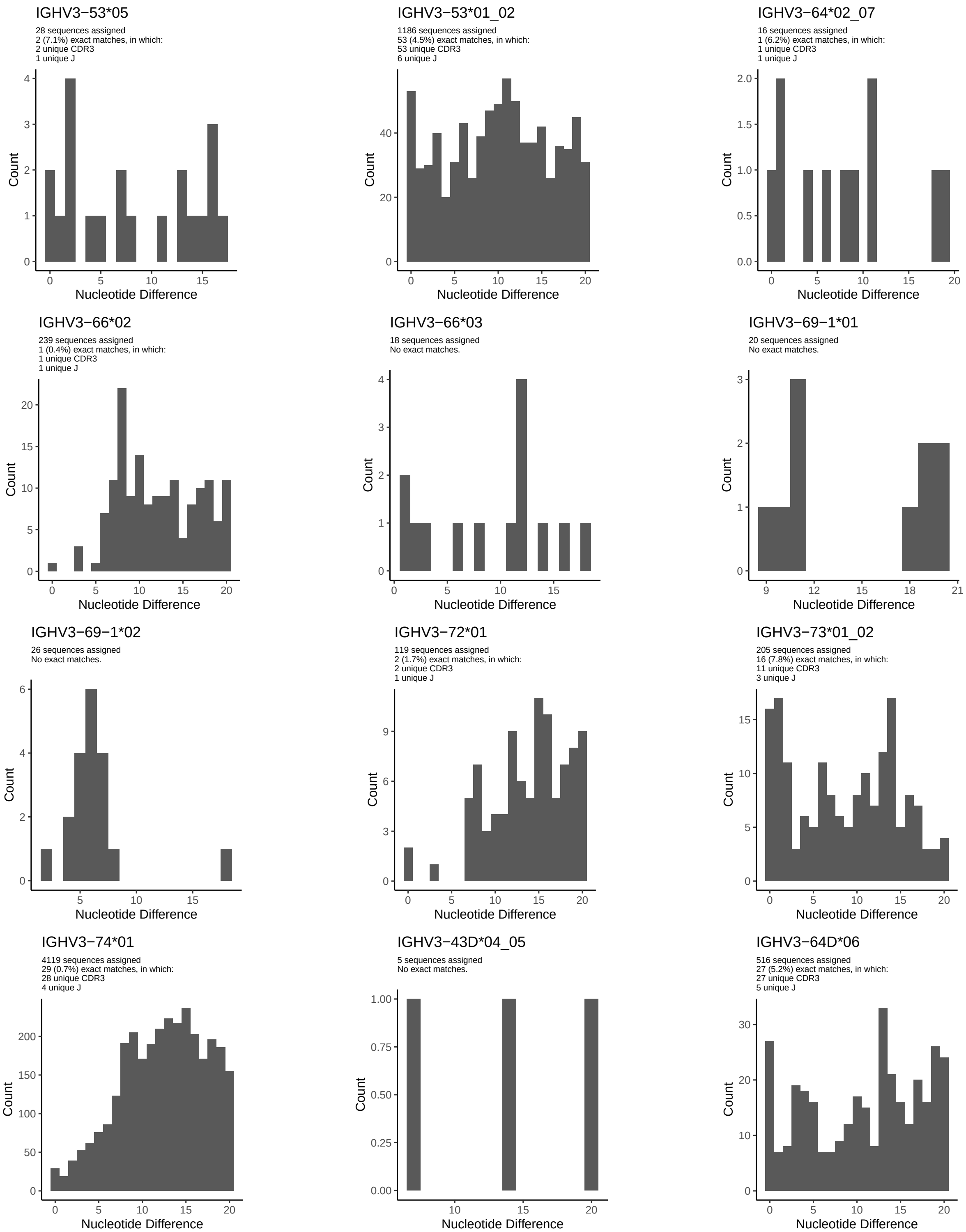
2 Variation from germline, in assignments to each allele



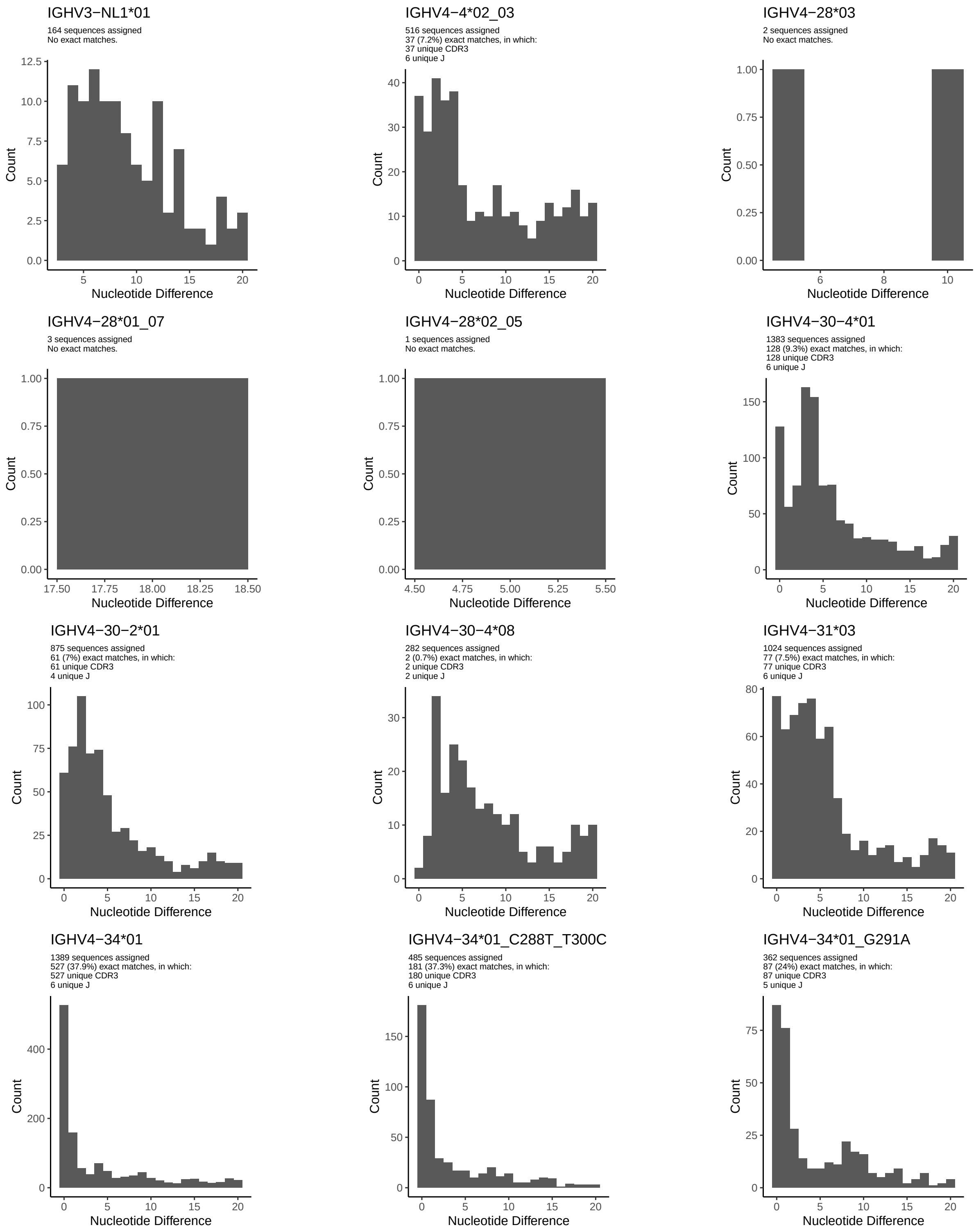

IGHV1−69*01_12_13
3097 sequences assigned
1170 (37.8%) exact matches, in which:
1158 unique CDR3
6 unique J
IGHV2−5*02
160 sequences assigned
30 (18.8%) exact matches, in which:
30 unique CDR3
4 unique J
IGHV2−26*01
20 sequences assigned
7 (35%) exact matches, in which:
7 unique CDR3
4 unique J
IGHV2−70*01
101 sequences assigned
19 (18.8%) exact matches, in which:
19 unique CDR3
3 unique J
IGHV3−7*01
2686 sequences assigned
83 (3.1%) exact matches, in which:
82 unique CDR3
6 unique J
IGHV3−7*03
2043 sequences assigned
66 (3.2%) exact matches, in which:
66 unique CDR3
5 unique J
IGHV3−9*01
1289 sequences assigned
74 (5.7%) exact matches, in which:
74 unique CDR3
4 unique J
IGHV3−11*01
647 sequences assigned
45 (7%) exact matches, in which:
45 unique CDR3
5 unique J
IGHV3−11*04
376 sequences assigned
22 (5.9%) exact matches, in which:
22 unique CDR3
3 unique J
IGHV3−11*05
151 sequences assigned
9 (6%) exact matches, in which:
9 unique CDR3
4 unique J
IGHV3−11*06
376 sequences assigned
38 (10.1%) exact matches, in which:
38 unique CDR3
4 unique J
IGHV3−13*01
60 sequences assigned
14 (23.3%) exact matches, in which:
14 unique CDR3
5 unique J
Count
Nucleotide Difference

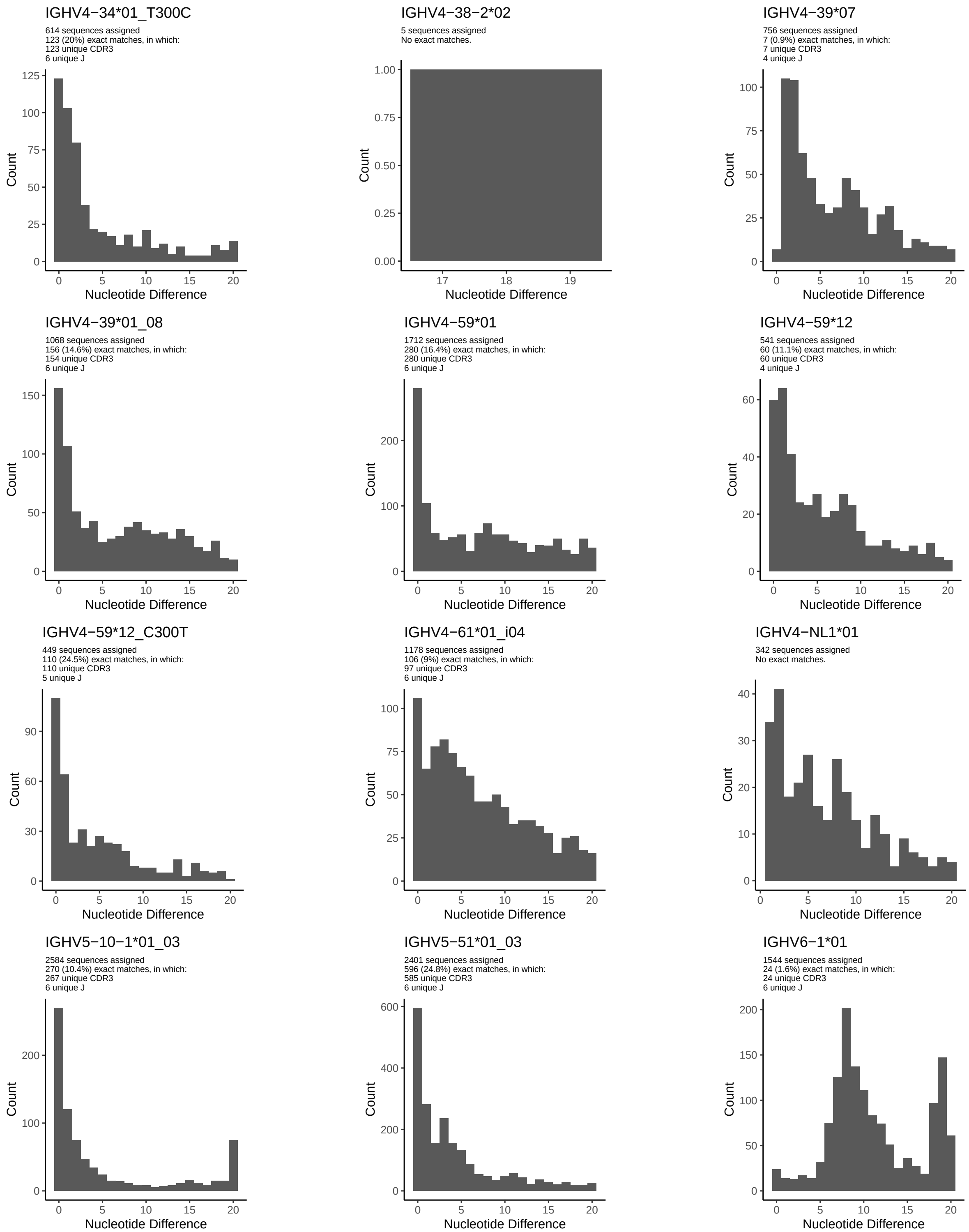

IGHV3−13*05
21 sequences assigned
5 (23.8%) exact matches, in which:
5 unique CDR3
3 unique J
IGHV3−13*06
30 sequences assigned
No exact matches.
IGHV3−15*01
398 sequences assigned
30 (7.5%) exact matches, in which:
30 unique CDR3
5 unique J
IGHV3−20*04
424 sequences assigned
3 (0.7%) exact matches, in which:
3 unique CDR3
2 unique J
IGHV3−21*01
2911 sequences assigned
325 (11.2%) exact matches, in which:
324 unique CDR3
6 unique J
IGHV3−23*03
2301 sequences assigned
139 (6%) exact matches, in which:
139 unique CDR3
6 unique J
IGHV3−23*01_04
5053 sequences assigned
114 (2.3%) exact matches, in which:
114 unique CDR3
6 unique J
IGHV3−30−3*01
3002 sequences assigned
837 (27.9%) exact matches, in which:
834 unique CDR3
6 unique J
IGHV3−30*03
764 sequences assigned
217 (28.4%) exact matches, in which:
216 unique CDR3
6 unique J
IGHV3−30*18
1182 sequences assigned
229 (19.4%) exact matches, in which:
227 unique CDR3
6 unique J
IGHV3−30−3*01_G317A
792 sequences assigned
211 (26.6%) exact matches, in which:
211 unique CDR3
6 unique J
IGHV3−30−3*01_T288C
1851 sequences assigned
167 (9%) exact matches, in which:
167 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV3−30*03_T288C
1164 sequences assigned
98 (8.4%) exact matches, in which:
97 unique CDR3
6 unique J
IGHV3−30*20_G75C
944 sequences assigned
164 (17.4%) exact matches, in which:
164 unique CDR3
5 unique J
IGHV3−33*01
671 sequences assigned
72 (10.7%) exact matches, in which:
72 unique CDR3
6 unique J
IGHV3−33*06
253 sequences assigned
6 (2.4%) exact matches, in which:
5 unique CDR3
3 unique J
IGHV3−33*08
520 sequences assigned
37 (7.1%) exact matches, in which:
37 unique CDR3
5 unique J
IGHV3−33*01_C288T
249 sequences assigned
65 (26.1%) exact matches, in which:
65 unique CDR3
5 unique J
IGHV3−43*01
98 sequences assigned
5 (5.1%) exact matches, in which:
5 unique CDR3
4 unique J
IGHV3−48*01
448 sequences assigned
16 (3.6%) exact matches, in which:
16 unique CDR3
4 unique J
IGHV3−48*02
600 sequences assigned
63 (10.5%) exact matches, in which:
63 unique CDR3
5 unique J
IGHV3−48*04
416 sequences assigned
9 (2.2%) exact matches, in which:
9 unique CDR3
2 unique J
IGHV3−49*04
15 sequences assigned
No exact matches.
IGHV3−49*03_05
67 sequences assigned
14 (20.9%) exact matches, in which:
14 unique CDR3
5 unique J
Count
Nucleotide Difference

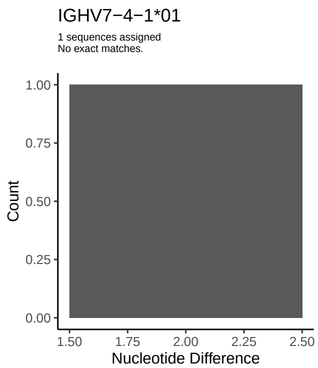


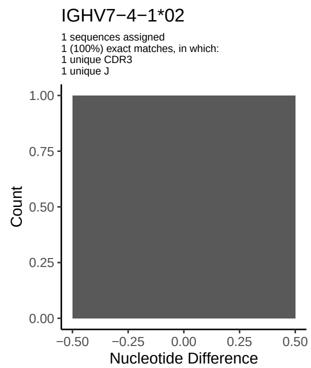
IGHV3−53*05
28 sequences assigned
2 (7.1%) exact matches, in which:
2 unique CDR3
1 unique J
IGHV3−53*01_02
1186 sequences assigned
53 (4.5%) exact matches, in which:
53 unique CDR3
6 unique J
IGHV3−64*02_07
16 sequences assigned
1 (6.2%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−66*02
239 sequences assigned
1 (0.4%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−66*03
18 sequences assigned
No exact matches.
IGHV3−69−1*01
20 sequences assigned
No exact matches.
IGHV3−69−1*02
26 sequences assigned
No exact matches.
IGHV3−72*01
119 sequences assigned
2 (1.7%) exact matches, in which:
2 unique CDR3
1 unique J
IGHV3−73*01_02
205 sequences assigned
16 (7.8%) exact matches, in which:
11 unique CDR3
3 unique J
IGHV3−74*01
4119 sequences assigned
29 (0.7%) exact matches, in which:
28 unique CDR3
4 unique J
IGHV3−43D*04_05
5 sequences assigned
No exact matches.
IGHV3−64D*06
516 sequences assigned
27 (5.2%) exact matches, in which:
27 unique CDR3
5 unique J
Count
Nucleotide Difference


IGHV3−NL1*01
164 sequences assigned
No exact matches.
IGHV4−4*02_03
516 sequences assigned
37 (7.2%) exact matches, in which:
37 unique CDR3
6 unique J
IGHV4−28*03
2 sequences assigned
No exact matches.
IGHV4−28*01_07
3 sequences assigned
No exact matches.
IGHV4−28*02_05
1 sequences assigned
No exact matches.
IGHV4−30−4*01
1383 sequences assigned
128 (9.3%) exact matches, in which:
128 unique CDR3
6 unique J
IGHV4−30−2*01
875 sequences assigned
61 (7%) exact matches, in which:
61 unique CDR3
4 unique J
IGHV4−30−4*08
282 sequences assigned
2 (0.7%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV4−31*03
1024 sequences assigned
77 (7.5%) exact matches, in which:
77 unique CDR3
6 unique J
IGHV4−34*01
1389 sequences assigned
527 (37.9%) exact matches, in which:
527 unique CDR3
6 unique J
IGHV4−34*01_C288T_T300C
485 sequences assigned
181 (37.3%) exact matches, in which:
180 unique CDR3
6 unique J
IGHV4−34*01_G291A
362 sequences assigned
87 (24%) exact matches, in which:
87 unique CDR3
5 unique J
Count
Nucleotide Difference

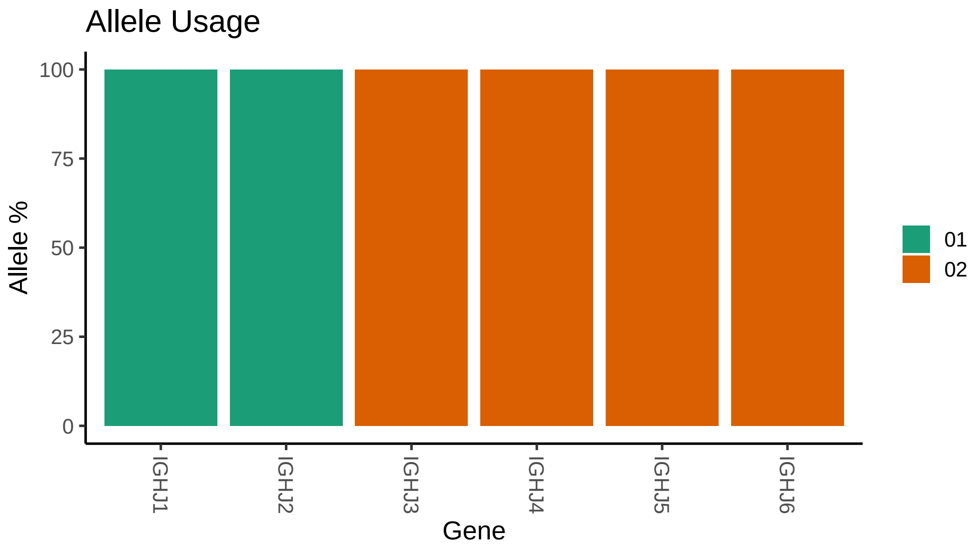


IGHV4−34*01_T300C
614 sequences assigned
123 (20%) exact matches, in which:
123 unique CDR3
6 unique J
IGHV4−38−2*02
5 sequences assigned
No exact matches.
IGHV4−39*07
756 sequences assigned
7 (0.9%) exact matches, in which:
7 unique CDR3
4 unique J
IGHV4−39*01_08
1068 sequences assigned
156 (14.6%) exact matches, in which:
154 unique CDR3
6 unique J
IGHV4−59*01
1712 sequences assigned
280 (16.4%) exact matches, in which:
280 unique CDR3
6 unique J
IGHV4−59*12
541 sequences assigned
60 (11.1%) exact matches, in which:
60 unique CDR3
4 unique J
IGHV4−59*12_C300T
449 sequences assigned
110 (24.5%) exact matches, in which:
110 unique CDR3
5 unique J
IGHV4−61*01_i04
1178 sequences assigned
106 (9%) exact matches, in which:
97 unique CDR3
6 unique J
IGHV4−NL1*01
342 sequences assigned
No exact matches.
IGHV5−10−1*01_03
2584 sequences assigned
270 (10.4%) exact matches, in which:
267 unique CDR3
6 unique J
IGHV5−51*01_03
2401 sequences assigned
596 (24.8%) exact matches, in which:
585 unique CDR3
6 unique J
IGHV6−1*01
1544 sequences assigned
24 (1.6%) exact matches, in which:
24 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV7−4−1*01
1 sequences assigned
No exact matches.
Count
1.00
0.75
0.50
0.25
0.00
1.50
1.75
2.00
2.25
2.50
Nucleotide Difference


IGHV7−4−1*02
1 sequences assigned
1 (100%) exact matches, in which:
1 unique CDR3
1 unique J
Count
1.00
0.75
0.50
0.25
0.00
−0.50
−0.25
0.00
0.25
0.50
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /work/jenkins/PRJNA491287/M64 028/M64 028/M64 028/M64 028/e5/9d6e826e70af812e31068b
##
## Germline reference file: /work/jenkins/PRJNA491287/M64 028/M64 028/M64 028/M64 028/e5/9d6e826e70af81
##
## Novel allele file: /work/jenkins/PRJNA491287/M64 028/M64 028/M64 028/M64 028/e5/9d6e826e70af812e3106
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 18 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 18 01_G276C_C291G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 18 01_T81A_T82G_A83G_T90C_C104G_G113C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 20_G75C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_G317A: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 01_C288T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_G291A: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 34 01_C288T_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
```